

TIRATH BHATHAWALA

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Education

New York University, Courant Institute of Mathematical Sciences

Aug 2026 – May 2028

Master of Science in Computer Science

New York, NY

SVKM's Dwarkadas J. Sanghvi College of Engineering

Nov 2022 – Jun 2026

B.Tech in Computer Engineering with Honors in Intelligent Computing (CGPA: 9.28)

Mumbai, India

Experience

BrandContext.ai

Mar 2025 – Aug 2025

AI Intern

Hybrid, Mumbai, India

- Developed an **AI-agentic system** powered by **dynamic graphs** and real-time factory operation databases to support supervisor workflows, detect anomalies, and dynamically incorporate worker-input special conditions.
- Evaluated and productionized computer-vision pipelines for retail supermarkets, fine-tuning **YOLOv11** multi-class detectors for Xiaomi (**0.87 mAP@50** across over 500 stores in India).
- Built legally-compliant **speaker-diarization** pipelines for sales-customer calls using open-source (**SpeechBrain**) and commercial (**ElevenLabs**) stacks to enable accurate speaker labeling and downstream analytics.

Indian Institute of Technology, Patna

Oct 2024 – May 2025

Research Intern

Remote, Patna, India

- Designed **MetaSearch**, a search-augmented, reasoning-based **AI Agent** for consensus resolution in peer review, combining disagreement detection and fact-grounded synthesis to automate meta-review decision-making.
- Developed a novel LLM-argumentation system for peer review, evaluating 900+ ICLR reviews using leading LLMs (**Llama 3.1**, **Mixtral**, **Gemini 2.0**, **GPT-4o**), achieving high inter-annotator agreement (**Cohen's Kappa = 0.834**).

Nvelop Technologies Oy

Aug 2024 – Feb 2025

AI Intern

Remote, Helsinki, Finland

- Devised **Retrieval-Augmented Generation (RAG)** workflows in **Azure AI PromptFlow** across 50+ RFPs.
- Optimized chunking strategies and top-p sampling parameters, reducing document retrieval **perplexity by 15%**.

Projects & Publications

AntiBug: Runtime-Aware Multimodal Agentic Program Repair | *LangGraph, Next.js, TypeScript*

April 2026

- Architected a **12-agent LangGraph pipeline** that ingests comprehensive multimodal telemetry, including resolved stack traces and DOM screenshots, for automated **root-cause diagnosis**, hypothesis validation, and patch synthesis.
- Evaluated on real-world production bugs, successfully fixing **46.38% of issues (3.5x better than GPT-4o baseline)**.
- Achieved a **72.50% human preference rate** during blind A/B patch evaluations conducted by professional developers.

FRAC-MAS: Conformal Multi-Agent Fracture Radiograph Analysis | *Grad-CAM, PyTorch*

March 2026

- Engineered a **multi-agent clinical decision support system** that integrates a **4-model vision ensemble** (MaxViT, RAD-DINO, YOLO, DenseNet) with specialized LLM agents and delivers verifiable, RAG-grounded reasoning.
- Implemented **conformal prediction** to generate statistically grounded differential diagnoses, achieving **92.0% empirical coverage** and a **96.4% detection rate** across external test sets like FracAtlas and MURA.

DigniFy: Multi-Modal Multilingual Hate Speech Detection | *HuggingFace, TensorFlow, FastAPI*

May 2025

- Engineered an **AI Agent** integrating modality-specific models, enabling unified hate speech detection through interactive orchestration across four LLMs (**DeepSeek R1**, **GPT-4o**, **Gemini 2.0**, **LLaMA 4.0**).
- Fine-tuned **DistilRoBERTa** for text analysis, implemented OCR for image extraction, and constructed an audio processing pipeline with STT-TTS, merging all modalities into a comprehensive video processing architecture.

Technical Skills

Languages: Python, C/C++, Java, Rust, SQL, JavaScript, HTML/CSS.

AI/ML Frameworks: PyTorch, TensorFlow, HuggingFace, OpenCV, scikit-learn, Transformers, Keras.

Tools/Cloud: Linux, Git, Docker, Azure AI, FastAPI.

Relevant Coursework: AI, ML, DL, NLP, Data Structures & Algorithms, Data Warehousing & Mining, Advanced DBMS.

Achievements

- Amazon ML Challenge 2024:** Secured Rank 292 out of 70,000+ participating teams nationwide.
- CodeBounty 2024:** 1st Place Winner in a college competitive programming contest out of 100+ registrations.
- Rajasthan Police Hackathon 2024:** Top 100 out of 1000+ teams (ML Lead).
- National AI Examination (GATE DS&AI):** Achieved All India Rank 499 (Top 0.8%) out of 60,000+ candidates.